

CLINICAL INVESTIGATION

Eye

LOCAL TUMOR CONTROL AFTER ^{106}Ru BRACHYTHERAPY OF CHOROIDAL MELANOMA

BERTIL DAMATO, PH.D., F.R.C.OPHTH.,* IMRAN PATEL, M.SC.,[†] IAN R. CAMPBELL, B.SC.,[‡]
HELEN M. MAYLES, M.SC.,[†] AND R. DOUGLAS ERRINGTON, F.R.C.R.[§]

*Ocular Oncology Service, St. Paul's Eye Unit, Royal Liverpool University Hospital, Liverpool, United Kingdom; Departments of
[†]Physics and [§]Radiotherapy, Clatterbridge Centre for Oncology, Bebington, Wirral, United Kingdom; [‡]IC Statistical Services,
Wirral, United Kingdom

Purpose: To report on local tumor control after ^{106}Ru brachytherapy for choroidal melanoma.

Methods and Materials: A total of 458 patients with choroidal melanoma were treated at a single institution between January 1993 and December 2001. The tumors had a median longest basal dimension of 10.6 mm and a median height of 3.2 mm. The brachytherapy was administered using a 15- or 20-mm plaque. For posterior tumors, the plaque was positioned eccentrically with its posterior edge aligned with the posterior tumor margin to reduce the radiation dose to the optic disk and fovea. A minimal scleral dose sufficient to cause visible choroidal atrophy provided a permanent ophthalmoscopic record of the distribution of choroidal irradiation. If radiotherapy to the posterior tumor was uncertain, adjunctive transpupillary thermotherapy was administered 6 months postoperatively.

Results: The actuarial rates of tumor recurrence were 1%, 2%, and 3% at 2, 5, and 7 years, respectively. Local tumor recurrence correlated with the longest basal tumor dimension (Cox univariate analysis, $p = 0.02$, risk ratio 1.41, 95% confidence interval 1.06–1.88). Seven of the nine eyes with recurrent tumor were salvaged with additional conservative therapy.

Conclusion: The low rate of local tumor recurrence suggests that ruthenium plaque radiotherapy is effective with good case selection and if special measures are taken to ensure that the plaque is positioned correctly.
© 2005 Elsevier Inc.

Uveal melanoma, ^{106}Ru , Plaque brachytherapy, Local control.

INTRODUCTION

Enucleation for uveal melanoma has largely been replaced by therapies aimed at destroying the primary tumor while conserving the eye with as much useful vision as possible. Such “conservative” modalities include plaque brachytherapy, proton beam radiotherapy, stereotactic radiotherapy, transpupillary thermotherapy, cryotherapy, transscleral local resection, and transretinal endoresection (1).

^{106}Ru plaques emit β -radiation so that tumors up to 5.4 mm in height can be treated reliably. Several investigators have independently published the results of plaque radiotherapy, administered alone or in combination with transpupillary thermotherapy (2–14). The survival rates have not been significantly different from those reported after enucleation (15–17). The conventional practice is to position the plaque centrally over the tumor so that it overlaps the visible tumor margins by at least 2 mm in all directions (10,

18). The wide radiation safety margin inevitably increases ocular morbidity, especially when the tumor extends close to optic disk or fovea. Since 1993, we have, whenever possible, treated such posterior choroidal melanomas with the posterior edge of the plaque aligned with the posterior tumor margin. This practice developed from the experience of one of us (B.D.) in treating juxtapapillary tumors with an anteriorly positioned ^{106}Ru plaque combined with low-energy, long-duration photocoagulation (19) to the posterior part of the tumor. To ensure accurate positioning of the plaque in relation to the tumor, any overlying extraocular muscles were routinely disinserted and both plaque and tumor were localized by indentation and transillumination. Furthermore, a minimal scleral radiation dose of 300 Gy was delivered so that choroidal atrophy developed, which enabled correct positioning of the plaque to be verified ophthalmoscopically during subsequent surveillance of the patient.

Reprint requests to: Bertil Damato, Ph.D., F.R.C.Ophth, St. Paul's Eye Unit, Royal Liverpool University Hospital, Prescot St., Liverpool L7 8XP, UK. Tel: (+44) 151-706-3973; Fax: (+44) 151-706-5436; E-mail: Bertil@damato.co.uk

The Liverpool Ocular Oncology Centre is funded by the National Supraregional Committee Advisory Group (NSCAG) of the

Department of Health of England, UK.

Acknowledgments—We are grateful to Gary Cheetham, for database support, and the National Specialist Commissioning Advisory Group (NSCAG) for funding our service.

Received Jan 10, 2005, and in revised form Feb 14, 2005.
Accepted for publication Feb 15, 2005.